RANGES AND INSEPARABILITY IN RECURSION CATEGORIES

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Abstract. In a P-recursion category with binary coproducts and ranges, the domain of a weak Turing morphism applied to the diagonal has no complement when the identity of its underlying object is nonzero. We also construct recursively inseparable domains using a range functor to a split restriction category satisfying weak choice, in which inverse images of domains along every morphism preserve binary joins. This construction requires each target morphism selecting between the images of a complemented source domain and its complement to have a preimage under the functor. Joins of source morphisms with disjoint domains suffice for this requirement. A counterexample satisfies the remaining hypotheses, but the constructed pair is separable and the required preimage does not exist.

1. Introduction

Di Paola and Heller developed recursion categories as an algebraic setting for recursion theory, with the aim of obtaining a categorical formulation of incompleteness [DPH87]. Morphisms represent partial computations, and restriction idempotents represent their domains. Rosolini's P-categories provide a more general setting for these constructions [Ros86, Ros88], used in subsequent existence and relativization theorems [Hel90, Ste93, Len04].

We study two diagonal arguments in this setting. The first produces a domain with no complement in a P-recursion category with binary coproducts and ranges, provided that the identity of the underlying object is nonzero (Theorem 3.1). The second produces a recursively inseparable pair of domains. Such a pair consists of two disjoint domains for which no complemented domain contains the first while its complement contains the second.

The inseparability argument uses a range functor, which preserves the domain and range operations, to a split restriction category satisfying weak choice, in which inverse images of domains along every morphism preserve binary joins. Under the target hypotheses of Theorem 4.1, every complemented source domain determines a target morphism that agrees with one coproduct injection on the image of the domain and with the other on the image of its complement. The lifting hypothesis requires that this morphism be the image of a source morphism. Joins of source morphisms with disjoint domains suffice for this hypothesis (Proposition 4.1.1). The counterexample in Section 4.5 satisfies the other hypotheses of the theorem, but its diagonal pair is separable; the target morphism has no source preimage.

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Examples defined by arithmetic provability motivate the use of a comparison functor. Let T be a consistent recursively enumerable extension of Peano arithmetic (PA). In the category S_T, morphisms are equivalence classes of program indices: two indices are equivalent precisely when T proves a single universally quantified statement asserting that the programs have the same domain and values [DPM91, p. 646]. Montagna's category S is the case $T = \mathrm{PA}$ [Mon89, p. 105]. The proposed system S'_T, with $S' = S'_{\mathrm{PA}}$, instead uses separate proofs of equality at each standard input. Pointwise provable equality is not preserved by composition, so these proposed quotients do not define categories under that composition [Len26, Corollary 2].

Di Paola and Montagna prove that S_T is not locally connected for any such T [DPM91, Corollary 4.4]; if T is not Σ^0_1-sound, it has no nonzero connected domains [DPM91, Theorem 4.3(2)]. Heller's existence theorem uses local connectedness [Hel90]. These examples therefore motivate extending that theorem beyond locally connected categories, as proposed in [DPM91] and developed using range functors in [Len04].

Section 2 introduces restriction categories, near products, and Turing morphisms. Section 3 develops the range calculus needed for the noncomplementation theorem and proves that theorem. Section 4 constructs the target morphisms, states the lifting hypothesis, and proves the inseparability theorem, its sufficient condition, and the counterexample.

2. Restriction categories and recursion categories

2.1. **Restriction categories.** We write gf for the composite that applies f first and then g. A morphism $e : X \to X$ is *idempotent* if $e^2 = e$.

A *restriction category* is a category $\mathbf{C}$ with an operation assigning to each morphism $f : X \to Y$ an endomorphism $\overline{f} : X \to X$, subject to the following axioms whenever the composites are defined [CL02, CL03].

R1. $f\overline{f} = f$.
R2. $\overline{f}\,\overline{g} = \overline{g}\,\overline{f}$.
R3. $\overline{g\overline{f}} = \overline{g}\,\overline{f}$.
R4. $\overline{g}f = f\overline{gf}$.

These axioms imply that $\overline{f}$ is idempotent and that $\overline{gf} = \overline{\overline{g}f}$. The endomorphisms $\overline{f}$ are called *restriction idempotents*, or *domains*. For an object X, write $\mathrm{Dom}(X)$ for its domains. They form a meet-semilattice: $\alpha \leq \beta$ means $\alpha\beta = \alpha$, and $\alpha \wedge \beta = \alpha\beta$. A morphism $f : X \to Y$ is *total* if $\overline{f} = 1_X$; total morphisms form a subcategory $\mathbf{C}_T$.

For parallel morphisms $f, g : X \to Y$, the *restriction order* is defined by $f \leq g$ if $f = g\overline{f}$. It restricts to the order just defined on domains. A join $f \vee g$, when it exists, is the least upper bound in this order.

A *system of zero morphisms* consists of morphisms $0_{X,Y} : X \to Y$ such that $g0_{X,Y}f = 0_{W,Z}$ for all $f : W \to X$ and $g : Y \to Z$. Such a system is unique when it exists. We write $0_X = 0_{X,X}$ and use *restriction zeros*, which satisfy $\overline{0_{X,Y}} = 0_X$. Domains α, β on X are *disjoint* if $\alpha\beta = 0_X$; parallel morphisms f, g have *disjoint domains* if $\overline{f}\,\overline{g} = 0_X$. A morphism $f : X \to Y$ is *weakly total* if $fg = 0$ implies $g = 0$ for every $g : Z \to X$.

A restriction category is *split* if each domain $\varepsilon : X \to X$ has a splitting $\varepsilon = me$, where $e : X \to Z$, $m : Z \to X$, and $em = 1_Z$. The map m is then a total monomorphism.

Binary *restriction coproducts* are categorical coproducts with total injections $i_0 : X \to X + Y$ and $i_1 : Y \to X + Y$. For morphisms $f : X \to Z$ and $g : Y \to Z$, write $[f, g] : X + Y \to Z$ for their copairing. The *codiagonal* is $\nabla_X = [1_X, 1_X] : X + X \to X$; thus $[f, g] = \nabla_Z(f + g)$. Restrictions satisfy $\overline{f + g} = \overline{f} + \overline{g}$.

Cockett and Lack's restriction categories [CL02, CL03], equivalently Stefani's RDP-categories [Ste02], describe domains by this unary operation. We next recall the near products and diagonals needed to define Turing morphisms.

2.2. ***P*-categories.** Let $\mathbf{C}$ be a category. The *diagonal functor* $\Delta_{\mathbf{C}} : \mathbf{C} \to \mathbf{C} \times \mathbf{C}$ is given on objects by $A \mapsto (A, A)$ and on morphisms by $f \mapsto (f, f)$.

A *P-category* consists of a category $\mathbf{C}$ together with a bifunctor $\times : \mathbf{C} \times \mathbf{C} \to \mathbf{C}$, called a *near product*, a natural transformation $\Delta : 1_{\mathbf{C}} \to \times \circ \Delta_{\mathbf{C}}$, and for each object X of $\mathbf{C}$, natural transformations

$$p_{0(\ ),X} : (\ ) \times X \to 1_{\mathbf{C}}, \quad p_{1X,(\ )} : X \times (\ ) \to 1_{\mathbf{C}}$$

such that the conditions i), ii) and iii) below are satisfied [Ros86].

For morphisms $f : X \to Y$ and $g : X \to Z$, put $\langle f, g\rangle = (f \times g)\Delta_X : X \to Y \times Z$.

i) The following equations hold.

$$\begin{array}{ll} p_{0X,X}\Delta_X = 1_X = p_{1X,X}\Delta_X, & (p_{0X,Y} \times p_{1X,Y})\Delta_{X\times Y} = 1_{X\times Y}, \\ p_{0X,Y}(1_X \times p_{0Y,Z}) = p_{0X,Y\times Z}, & p_{0X,Z}(1_X \times p_{1Y,Z}) = p_{0X,Y\times Z}, \\ p_{1X,Z}(p_{0X,Y} \times 1_Z) = p_{1X\times Y,Z}, & p_{1Y,Z}(p_{1X,Y} \times 1_Z) = p_{1X\times Y,Z}. \end{array}$$

ii) There is an isomorphism of functors $\mathrm{ass}_\times : ((- \times -) \times -) \Rightarrow (- \times (- \times -))$ from $\mathbf{C}^3$ to $\mathbf{C}$, whose component $\mathrm{ass}_{\times\, X,Y,Z}$ is given by

$$\langle p_{0X,Y}p_{0X\times Y,Z}, \langle p_{1X,Y}p_{0X\times Y,Z}, p_{1X\times Y,Z}\rangle\rangle : (X \times Y) \times Z \to X \times (Y \times Z),$$

iii) Let T be the endofunctor on $\mathbf{C} \times \mathbf{C}$ given by $(X, Y) \mapsto (Y, X)$. There is an isomorphism $\mathrm{tr}_\times : \times \Rightarrow \times \circ T$ of functors from $\mathbf{C}^2$ to $\mathbf{C}$, with component $\mathrm{tr}_{\times X,Y} = \langle p_{1X,Y}, p_{0X,Y}\rangle : X \times Y \to Y \times X$. The isomorphisms $\mathrm{ass}_\times$ and $\mathrm{tr}_\times$ are natural in each of their component variables.

The induced restriction of a morphism $\varphi : X \to Y$ is $\overline{\varphi} = p_0(1_X \times \varphi)\Delta_X = p_1(\varphi \times 1_X)\Delta_X$. This operation satisfies R1–R4 [DPH87, Ros86].

The near product restricts to a categorical product on $\mathbf{C}_T$ [Ros86].

In general, a P-category is equivalent to one obtained from a category with products, following a procedure analogous to that by which the category **Pfn** of sets and partial functions can be obtained from the category **Set** of sets [Ros88, Theorem 1.2].

The projection p_0 is natural in the first variable, and p_1 is natural in the second. Thus $p_0(\varphi \times 1_X) = \varphi p_0$, whereas $p_1(\varphi \times 1_X) = 1_X p_1$ can fail: the right-hand side is total, while the left-hand side need not be. In general, the projections $p_{i,X,Y}$ of a P-category $\mathbf{C}$ are natural in both variables if and only if for any morphisms $\varphi_i : Y \to X_i$ $(i = 0, 1)$ in $\mathbf{C}$, $p_i\langle\varphi_0, \varphi_1\rangle = \varphi_i$. Cockett and Lack observe that P-categories are "symmetric monoidal categories in which each object has a monoidal natural cocommutative coassociative comultiplication (and possibly an unnatural counit)" [CL02].

A P-category *has coproducts* if its induced restriction category has binary restriction coproducts. It is *prodominical* if it has zero morphisms and $\phi \times 0_{C,D} = 0_{A\times C,B\times D}$ for every morphism $\phi : A \to B$ [Hel90]. We assume throughout that our P-categories are prodominical. Their zero domains are the least elements of $\mathrm{Dom}(X)$.

A prodominical category is *dominical* if every weakly total morphism is total and $f \times g = 0$ implies $f = 0$ or $g = 0$ [Len04, p. 36].

A category is *isotypical* if any two of its objects are isomorphic.

2.3. **Turing morphisms and P-recursion categories.** A *Turing morphism* in a prodominical isotypical P-category is a morphism $\tau : W \times X \to Y$ such that for any $\phi : V \times X \to Y$ there exists a total $g : V \to W$, such that the following diagram commutes.

$$\begin{array}{ccc} V \times X & \xrightarrow{g \times 1_X} & W \times X \\ & \searrow^{\phi} & \downarrow{\scriptstyle \tau} \\ & & Y \end{array}$$

A *P-recursion category* (or, simply, a *recursion category*) is a prodominical isotypical P-category which contains a Turing morphism. Previously, the term *recursion category* was used for dominical P-recursion categories; for us a recursion category is always a P-recursion category. The prototypical example of a Turing morphism $t : X \times X \to X$ is the universal partial recursive function [Odi89, Theorem II.1.8, page 132]. Examples of Turing morphisms and recursion categories are given in [DPH87, Mon89, Hel90, DPM91, Ste93, Len04].

A morphism $t : X \times X \to X$ is *weak Turing* if, for every $f : X \to X$, there is a total morphism $e : X \to X$ such that

$$\begin{array}{ccc} X \times X & \xrightarrow{e \times 1_X} & X \times X \\ {\scriptstyle p_{1,X,X}}\downarrow & & \downarrow{\scriptstyle t} \\ X & \xrightarrow[f]{} & X \end{array} \tag{1}$$

commutes. Such an e is a *t-index* of f. The morphism t is *idempotent weak Turing* if every f has an idempotent t-index. Every Turing morphism of type $X \times X \to X$ is weak Turing; conversely, a Turing morphism can be obtained from a weak Turing morphism [Hel90, Lemma 9.1].

For every total t-index e of f, precomposing (1) with $\langle 1_X, e \rangle$ gives

$$t\Delta_X e = fe. \tag{2}$$

Indeed, $(e \times 1_X)\langle 1_X, e\rangle = \Delta_X e$ and $p_{1,X,X}\langle 1_X, e\rangle = e$. This identity will be used in both diagonal arguments.

Proposition 2.0.1. *Every P-recursion category $\mathbf{C}$ has an idempotent weak Turing morphism t, together with a mapping N such that for every t-index e of φ in $\mathbf{C}$, $N(e)$ is an idempotent t-index of φ.*

Proof. Choose a Turing morphism $\tau : X \times X \to X$, put $Y = X \times X$, and choose an isomorphism $a : Y \to X$. Let $q_0, q_1 : Y \to X$ be the projections. The morphism $t = a^{-1}\tau(q_1 \times a) : Y \times Y \to Y$ is Turing: given $f : V \times Y \to Y$, apply the parameterization property of τ to $af(1_V \times a^{-1}) : V \times X \to X$, and duplicate the resulting parameter map to obtain a map $V \to Y$.

For a total t-index $e : Y \to Y$, put $k = q_1 e \Delta_X : X \to X$ and $N(e) = \langle q_0, kq_0\rangle : Y \to Y$. Then $q_0 N(e) = q_0$, so $N(e)^2 = N(e)$. If e indexes $f : Y \to Y$, precomposing its index equation with $(\Delta_X q_0) \times 1_Y$ gives $a^{-1}\tau(kq_0 \times a) = fp_{1,Y,Y}$. The left-hand side is $t(N(e) \times 1_Y)$, so $N(e)$ is an idempotent t-index of f. □

3. Ranges and the diagonal domain

We first record the properties of ranges needed for the diagonal argument.

3.1. **Ranges.** In this subsection, ranges are defined in a restriction category with restriction zeros and binary restriction coproducts. The definition applies in particular to the P-categories under consideration.

For a morphism $\phi : X \to Y$ and $\varepsilon \in \mathrm{Dom}(Y)$, we write $\phi \prec \varepsilon$ if and only if $\varepsilon\phi = \phi$; we say that ε *receives* ϕ. If ε receives ϕ, and in addition, ε satisfies for all appropriate ψ, ψ', $\psi\phi = \psi'\phi$ implies $\psi\varepsilon = \psi'\varepsilon$, then ε is the least domain in $\mathrm{Dom}(Y)$ receiving ϕ. In this case we say that ϕ has range ε and we write $\mathrm{ran}\phi = \varepsilon$.

The relation $\prec$ agrees with $\leq$ on domains. It has the following transitivity property.

Proposition 3.0.1. *If $\phi : X \to Y$ is a morphism and β, γ are domains in Y such that $\phi \prec \beta$ and $\beta \leq \gamma$, then $\phi \prec \gamma$.*

Proof. We have $\phi = \beta\phi = (\beta\gamma)\phi = \gamma(\beta\phi) = \gamma\phi$, so that $\phi \prec \gamma$. □

A restriction category has ranges if every morphism has a range in the sense above. For a P-category $\mathbf{C}$, we require in addition that for morphisms ϕ, ψ of $\mathbf{C}$, $\mathrm{ran}(\phi \times \psi) = \mathrm{ran}\phi \times \mathrm{ran}\psi$, so that ranges are compatible with the near product. Such a category has a calculus of ranges [Hel90, DPH87].

In a restriction category with coproducts and ranges, the meet-semilattice $\mathrm{Dom}(X)$ becomes a distributive lattice if one defines the join by $\varepsilon \vee \delta = \mathrm{ran}[\varepsilon,\ \delta]$ for $\varepsilon, \delta \in \mathrm{Dom}(X)$ [DPH87, Hel90]. The next statement follows from the definition of range [DPH87, Proposition 5.4].

Proposition 3.0.2. *If ε is a domain in $\mathrm{Dom}(Y)$ and $\varphi : X \to Y$ is any morphism, then $\mathrm{ran}(\varepsilon\varphi) = \varepsilon\mathrm{ran}(\varphi)$.*

Proof. Put $r = \mathrm{ran}(\varphi)$ and $s = \mathrm{ran}(\varepsilon\varphi)$. The domain εr receives $\varepsilon\varphi$, so $s \leq \varepsilon r$. Conversely, $s\varepsilon\varphi = \varepsilon\varphi$; the equality property of r gives $s\varepsilon r = \varepsilon r$. Hence $\varepsilon r \leq s$. □

The least-receiving property shows that $\mathrm{ran}[\alpha, \beta]$ is the join of α and β. The proposition gives distributivity, since $\varepsilon(\alpha \vee \beta) = \mathrm{ran}[\varepsilon\alpha, \varepsilon\beta] = \varepsilon\alpha \vee \varepsilon\beta$.

A domain $\gamma \in \mathrm{Dom}(X)$ is *complemented* if there is a domain $\gamma^\perp$ with $\gamma \vee \gamma^\perp = 1_X$ and $\gamma\gamma^\perp = 0_X$. Distributivity makes this complement unique.

The next fact follows from the preceding.

Proposition 3.0.3. *For any morphisms $\varphi : X \to Y, \psi : Y \to Z$, $\psi\varphi = 0$ if and only if $\overline{\psi}\mathrm{ran}(\varphi) = 0$.*

Proof. If $\overline{\psi}\mathrm{ran}(\varphi) = 0$, then $\psi\varphi = \psi\overline{\psi}\mathrm{ran}(\varphi)\varphi = 0$. Conversely, if $\psi\varphi = 0$, then $0 = \overline{\psi\varphi} = \overline{\overline{\psi}\varphi}$ by the restriction identity $\overline{gf} = \overline{\overline{g}f}$, which implies that $\overline{\psi}\varphi = 0$. Take ranges and apply the preceding proposition. □

For a morphism $f : X \to Y$, define its *inverse image* and *image* on domains by $f^*\beta = \overline{\beta f}$ and $f_*\alpha = \mathrm{ran}(f\alpha)$. Inverse image preserves meets, and image preserves joins. In a P-category with coproducts and ranges, inverse image also preserves binary joins [DPH87, Hel90]. Whenever the target is a restriction category without a specified near product, we require this last property explicitly. The following fact will be used below [DPH87, Cor. 5.6 (iii)].

Proposition 3.0.4. *Let $\phi : X \to Y$ be a morphism, and let $\alpha \in \mathrm{Dom}(X)$ and $\beta \in \mathrm{Dom}(Y)$, then $\phi_*(\alpha)\beta = \phi_*(\alpha\phi^*\beta)$.*

Proof. By R4 and commutativity of domains, $\phi\alpha\phi^*\beta = \beta\phi\alpha$. Take ranges and apply Proposition 3.0.2. □

For ϕ total we have that ϕ_* is left adjoint to ϕ^* [Hel90].

Proposition 3.0.5. *If $\phi : X \to Y$ is total in a restriction category with ranges, then for each $\alpha \in \mathrm{Dom}(X)$ and for each $\beta \in \mathrm{Dom}(Y)$, $\phi_*\alpha \leq \beta$ if and only if $\alpha \leq \phi^*\beta$.*

Adjointness follows from the next proposition.

Proposition 3.0.6. *If $\phi : X \to Y$ is a morphism in a restriction category $\mathbf{C}$ with ranges, then for each $\alpha \in \mathrm{Dom}(X)$ and for each $\beta \in \mathrm{Dom}(Y)$,*
a) $\phi_*\alpha \leq \beta$ implies $\alpha\phi^*\beta = \alpha\overline{\phi}$; and
b) $\alpha \leq \phi^*\beta$ implies $\phi_*\alpha \leq \beta$.

Proof. For a), the condition implies $\beta\mathrm{ran}(\phi\alpha) = \mathrm{ran}(\phi\alpha)$. Composing on the right with $\phi\alpha$, we have $\beta\phi\alpha = \phi\alpha$. Taking restrictions gives $\overline{\beta\phi}\alpha = \overline{\beta\phi\alpha} = \overline{\phi\alpha} = \overline{\phi}\alpha$.
Assertion b) follows from Proposition 3.0.4. □

3.2. **A domain with no complement.**

Theorem 3.1. *Let $\mathbf{C}$ be a P-recursion category with binary coproducts and ranges. If $t : X \times X \to X$ is weak Turing and $1_X \neq 0_X$, then $\overline{t\Delta_X}$ is not complemented.*

Proof. Put $h = t\Delta_X$ and $\kappa = \overline{h}$. Suppose that a domain ε complements κ, and choose a total t-index $e : X \to X$ of ε. Equation (2) gives $he = \varepsilon e$, and hence $e^*\kappa = \overline{he} = \overline{\varepsilon e} = e^*\varepsilon$. Since e is total and inverse image preserves meets and binary joins, these equal domains have meet 0_X and join 1_X. Therefore $1_X = 0_X$, a contradiction. □

A domain $\delta \in \mathrm{Dom}(X)$ is a *quasicomplement* of $\varepsilon \in \mathrm{Dom}(X)$ if $\delta\varepsilon = 0_X$ and, for every $\phi : Y \to X$, equality $\varepsilon\phi = \delta\phi$ implies $\phi = 0$. Di Paola and Heller proved that in a dominical recursion category the diagonal domain has no quasicomplement [DPH87, Corollary 8.6]. Complements and quasicomplements coincide in a dominical category.

Proposition 3.1.1. *In a P-category with a system of zeros, binary coproducts and ranges, the following assertions are equivalent.*
(i) Weakly total morphisms are total.
(ii) [DPH87, Proposition 5.3] *If $\varepsilon \in \mathrm{Dom}(X)$ satisfies $\varepsilon \neq 1_X$, there exists $\delta \in \mathrm{Dom}(X)$ with $\delta \neq 0$ and $\varepsilon\delta = 0$.*
(iii) [DPH87, Proposition 7.3] *Complements and quasicomplements coincide.*

Rosolini has written in [Ros88, page 313] that in view of his representation Theorem 2.8 (*loc. cit.*), all the results of section 5 of [DPH87] "...can be generalized to P-categories with ranges stable under products by checking that they hold in a P-category of the form $\mathcal{M}$–Ptl($\mathbf{A}$)." The ranges in S are defined by existential quantification of graph formulas [Mon89, p. 111] and are stable under products, since the two coordinate variables can be quantified independently. The preceding proposition, together with Montagna's weakly total non-total morphism in S [Mon89, Example 1, p. 112], shows that Proposition 5.3 of [DPH87] does not extend to this generality. Theorem 3.1 gives the noncomplementation result directly.

4. Recursively inseparable domains

Let $\mathbf{C}$ be a P-recursion category with coproducts and ranges. A disjoint pair (α, β) in $\mathrm{Dom}(X)$ is *recursively separable* if there is a complemented domain $\gamma \in \mathrm{Dom}(X)$ with $\alpha \leq \gamma$ and $\beta \leq \gamma^\perp$. It is *recursively inseparable* if no such γ exists.

4.1. **Connected domains.** Connected domains are a surrogate in P-categories for points; they first appeared in [Hel90] and subsequently in [DPM91, Len04]. A domain α is *connected* if every decomposition $\alpha = \beta \vee \gamma$ with $\beta\gamma = 0$ has $\beta = 0$ or $\gamma = 0$. The zero domain is connected. A restriction category with ranges and binary restriction coproducts is *locally connected* if, for all $f, g : X \to Y$, equality $f\alpha = g\alpha$ for every connected domain α on X implies $f = g$ [Hel90]. When inverse images preserve binary joins, total morphisms preserve connected domains under direct image. This gives the following alternatives for a morphism into a coproduct.

Proposition 4.0.1. *Let* $\mathbf{C}$ *be a restriction category with ranges, restriction zeros, and binary restriction coproducts, in which inverse images of domains preserve binary joins. Suppose that* $\alpha \in \mathrm{Dom}(X)$ *is connected.*
(i) If $f : X \to Y$ *is total, then* $f_*\alpha$ *is connected.*
(ii) If $f : X \to Y + Z$ *is total, then* $f_*\alpha \leq 1_Y + 0_Z$ *or* $f_*\alpha \leq 0_Y + 1_Z$.
(iii) If $f : X \to Y + Z$ *is total, then* $f\alpha \prec 1_Y + 0_Z$ *or* $f\alpha \prec 0_Y + 1_Z$.

Proof. The proof of (i) uses adjointness and Proposition 3.0.4. Suppose that $f_*\alpha = \beta \vee \gamma$ with $\beta\gamma = 0$. Since f is total, by adjointness, $\alpha \leq f^*(\beta \vee \gamma) = f^*\beta \vee f^*\gamma$; equivalently, $\alpha = \alpha f^*\beta \vee \alpha f^*\gamma$. We have that $\alpha(f^*\beta)\alpha(f^*\gamma) = \alpha f^*(\beta\gamma) = \alpha\overline{0f} = 0$. Therefore, either $\alpha f^*\beta = 0$ or $\alpha f^*\gamma = 0$. If $\alpha f^*\beta = 0$, then $0 = f_*(\alpha f^*\beta) = f_*(\alpha)\beta = (\beta \vee \gamma)\beta = \beta$. Similarly, if $\alpha f^*\gamma = 0$, $\gamma = 0$.

For (ii), note that $f_*\alpha = f_*\alpha \wedge (1_Y + 1_Z) = f_*\alpha \wedge ((1_Y + 0) \vee (0 + 1_Z)) = (f_*\alpha \wedge (1_Y + 0)) \vee (f_*\alpha \wedge (0 + 1_Z))$; the conclusion follows from (i).

For (iii), note that $f_*\alpha$ is a range, so that $f\alpha \prec f_*\alpha$; i.e., $(f_*\alpha)f\alpha = f\alpha$. The result follows from (ii) and Proposition 3.0.1. □

4.2. **Range functors.** Let $\mathbf{C}$ and $\mathbf{D}$ be restriction categories with ranges, restriction zeros, and binary restriction coproducts. A *range functor* $F : \mathbf{C} \to \mathbf{D}$ preserves restrictions, ranges, zeros, and binary coproducts. It preserves total morphisms, meets and joins of domains, and complements. For objects X, Y of $\mathbf{C}$, the canonical coproduct comparison is the isomorphism

$$\chi_{X,Y} = [Fi_0, Fi_1] : FX + FY \longrightarrow F(X + Y).$$

For P-categories, preservation of the near product is not required; examples occur in categories of coalgebras [Len04].

The *kernel* of F is $\ker F = \{f : X \to Y \mid Ff = 0_{FX,FY}\}$; it is closed under composition on either side.

Proposition 4.0.2. *If* $F : \mathbf{C} \to \mathbf{D}$ *is a range functor and if* $\mathbf{D}$ *is locally connected, then for each object* X *in* $\mathbf{C}$ *such that* $1_X \notin \ker F$*, there exists a nonzero connected domain in* $\mathrm{Dom}(FX)$.

Proof. Suppose that 0 is the only connected domain in $\mathrm{Dom}(FX)$. Then $1_{FX}0 = 0$, which implies that $F1_X = 1_{FX} = 0$, since $\mathbf{D}$ is locally connected. But then $1_X \in \ker F$, contradiction. □

4.3. **Sections and weak choice.** In a restriction category, a *section* of a morphism $\phi : X \to Y$ is a morphism $\sigma : Y \to X$ such that $\phi\sigma = \overline{\sigma}$ and $\phi\sigma\phi = \phi$. It follows that $\overline{\sigma} = \mathrm{ran}(\phi)$. A restriction category satisfies the *axiom of choice* if every morphism has a section. A morphism $\phi : X \to Y$ is a *partial monomorphism* if for all morphisms θ, θ', $\phi\theta = \phi\theta'$ implies $\overline{\phi}\theta = \overline{\phi}\theta'$. A restriction category satisfies the *weak axiom of choice*, or *weak choice*, if every partial monomorphism has a section.

To check that a partial recursive function ϕ is a partial monomorphism, it suffices to check that for all (total) recursive functions θ, θ', $\phi\theta = \phi\theta'$ implies $\overline{\phi}\theta = \overline{\phi}\theta'$. This follows from the fact that a recursively enumerable set is either empty or the image of a total recursive function. The following is a generalization.

Proposition 4.0.3. *Let* $\mathbf{C}$ *be a split restriction category, and let* $f : X \to Y$ *be a morphism in* $\mathbf{C}$*. Suppose that, for every object* W *of* $\mathbf{C}$ *and all total morphisms* $g, h : W \to X$*, the equality* $fg = fh$ *implies* $\overline{f}\, g = \overline{f}\, h$*. Then* f *is a partial monomorphism.*

Proof. Let $g, h : W \to X$ satisfy $fg = fh$. Split the common restriction $\varepsilon = \overline{fg} = \overline{fh}$ as $\varepsilon = me$, where $e : W \to Z$, $m : Z \to W$, and $em = 1_Z$. Since m is a total monomorphism, $\overline{fgm} = \overline{\varepsilon m} = \overline{m} = 1_Z$. Consequently fgm and fhm are total. Therefore gm and hm are total. By the hypothesis, $\overline{f}gm = \overline{f}hm$. Moreover,

$$\overline{f}g = \overline{f}g\overline{\overline{f}g} = \overline{f}g\overline{fg} = \overline{f}gme = \overline{f}hme = \overline{f}h\overline{fh} = \overline{f}h\overline{\overline{f}h} = \overline{f}h.$$

Therefore, f is a partial monomorphism. □

Proposition 4.0.4. *Let* $\mathbf{C}$ *be a restriction category with ranges, restriction zeros and binary restriction coproducts, in which inverse images of domains along every morphism preserve binary joins. Let* X *be an object of* $\mathbf{C}$*, and suppose that* $\alpha, \beta \in \mathrm{Dom}(X)$ *satisfy* $\alpha\beta = 0_X$*. For total morphisms* $\theta, \theta' : Y \to X + X$*, equality* $\nabla_X(\alpha + \beta)\theta = \nabla_X(\alpha + \beta)\theta'$ *implies* $(\alpha + \beta)\theta = (\alpha + \beta)\theta'$*.*

Proof. Put $p = \nabla_X(\alpha+\beta) : X+X \to X$, so that $\overline{p} = \alpha+\beta$. Let $i_0, i_1 : X \to X+X$ be the coproduct injections, and define

$$\rho_0 = [1_X, 0_X], \qquad \rho_1 = [0_X, 1_X] : X + X \to X,$$
$$\varepsilon_i = i_i\rho_i : X + X \to X + X \qquad (i = 0, 1).$$

The summand domains $\varepsilon_0, \varepsilon_1$ are disjoint and have join 1_{X+X}. Define domains on Y by

$$e_i = \theta^*\varepsilon_i, \qquad f_j = (\theta')^*\varepsilon_j, \qquad \delta_{ij} = e_i f_j \qquad (i, j \in \{0, 1\}).$$

Totality and inverse-image preservation give $e_0 \vee e_1 = 1_Y = f_0 \vee f_1$. Distributivity gives $\delta_{i0} \vee \delta_{i1} = e_i$, and hence the four domains δ_{ij} have join 1_Y.

For fixed $i, j \in \{0, 1\}$, put

$$x_{ij} = \rho_i\theta\delta_{ij} : Y \to X, \qquad y_{ij} = \rho_j\theta'\delta_{ij} : Y \to X.$$

The restriction identity $\varepsilon_i\theta = \theta\overline{\varepsilon_i\theta}$ and its analogue for θ' give

$$\theta\delta_{ij} = i_i x_{ij}, \qquad \theta'\delta_{ij} = i_j y_{ij}.$$

Write $a_0 = \alpha$ and $a_1 = \beta$. The hypothesis $p\theta = p\theta'$ gives the common morphism

$$z_{ij} = a_i x_{ij} = a_j y_{ij} : Y \to X.$$

It follows that

$$(\alpha + \beta)\theta\delta_{ij} = i_i z_{ij}, \qquad (\alpha + \beta)\theta'\delta_{ij} = i_j z_{ij}.$$

If $i = j$, these morphisms agree. If $i \neq j$, then $z_{ij} = a_i z_{ij} = a_i a_j y_{ij} = 0_{Y,X}$, so both morphisms are zero. Thus they agree for every pair i, j.

Form the copairing

$$q = [[[\delta_{00}, \delta_{01}], \delta_{10}], \delta_{11}] : ((Y + Y) + Y) + Y \longrightarrow Y.$$

A domain receives q precisely when it receives each δ_{ij}, so $\mathrm{ran}(q) = 1_Y$. The preceding equalities give $(\alpha + \beta)\theta q = (\alpha + \beta)\theta' q$. Range agreement therefore gives $(\alpha + \beta)\theta = (\alpha + \beta)\theta'$. □

Proposition 4.0.5. *Let* $\mathbf{C}$ *be a split restriction category with ranges, restriction zeros and binary restriction coproducts. Suppose that inverse images of domains along every morphism preserve binary joins and that weak choice holds. If* $\alpha, \beta \in \mathrm{Dom}(X)$ *satisfy* $\alpha\beta = 0_X$*, then* $p = \nabla_X(\alpha + \beta) : X + X \to X$ *has a unique section* $f : X \to X + X$*. Moreover,* $pf = \overline{f} = \mathrm{ran}(p) = \alpha \vee \beta$*.*

Proof. Propositions 4.0.4 and 4.0.3 show that p is a partial monomorphism; here $\overline{p} = \alpha + \beta$. Weak choice supplies a section f. By R4 and the section equations, $\overline{p}\, f = f\overline{pf} = f\overline{f} = f$. If f' is another section, then $pf = pf' = \mathrm{ran}(p)$; partial monicity yields $f = \overline{p}\, f = \overline{p}\, f' = f'$. Finally, $\mathrm{ran}(p) = \alpha \vee \beta$ by the definition of join. □

Proposition 4.0.6. *Let* $\mathbf{C}$ *satisfy the hypotheses of Proposition 4.0.5, and let* $\alpha \in \mathrm{Dom}(X)$ *be complemented. The section* $f : X \to X + X$ *of* $p = \nabla_X(\alpha^\perp + \alpha)$ *is total and satisfies* $f\alpha = i_1\alpha$ *and* $f\alpha^\perp = i_0\alpha^\perp$*.*

Proof. Proposition 4.0.5 gives $\overline{f} = 1_X$ and $pf = 1_X$. Since $p(f\alpha) = \alpha = p(i_1\alpha)$, partial monicity gives $\overline{p}\, f\alpha = \overline{p}\, i_1\alpha$. The left-hand side is $f\alpha$, and the right-hand side is $i_1\alpha$. The equation for $\alpha^\perp$ follows in the same way. □

4.4. **The inseparability theorem.** Di Paola and Heller proved the existence of recursively inseparable domains in a dominical recursion category $\mathbf{C}$ with coproducts, under the assumption that $\mathbf{C}$ satisfies the axiom of choice [DPH87, Theorem 8.15]. Rosolini removed the requirement that $\mathbf{C}$ be dominical and that products distribute over coproducts, provided domains split in $\mathbf{C}$ [Ros88].

Theorem 4.1. *Let* $\mathbf{C}$ *be a P-recursion category with binary coproducts and ranges. Let* $\mathbf{D}$ *be a split restriction category with ranges, restriction zeros and binary restriction coproducts, satisfying weak choice, in which inverse images of domains along every morphism preserve binary joins. Let* $F : \mathbf{C} \to \mathbf{D}$ *be a range functor,* $t : X \times X \to X$ *a weak Turing morphism in* $\mathbf{C}$ *with* $1_X \notin \ker F$*, and* $b : X \to X + X$ *an isomorphism.*

For each complemented $\gamma \in \mathrm{Dom}(X)$*, let* $s_\gamma : FX \to FX + FX$ *be the unique section of* $\nabla_{FX}(F\gamma^\perp + F\gamma)$*. Assume the lifting hypothesis: for every such* γ*, there exists a morphism* $d_\gamma : X \to X + X$ *in* $\mathbf{C}$ *such that* $Fd_\gamma = \chi_{X,X} s_\gamma$*. Then the domains*

$$\alpha = (bt\Delta_X)^*(1_X + 0_X), \qquad \beta = (bt\Delta_X)^*(0_X + 1_X)$$

form a recursively inseparable pair in $\mathrm{Dom}(X)$*.*

Proof. Put $h = bt\Delta_X : X \to X + X$. Since inverse image preserves meets and zero, $\alpha\beta = h^*((1_X + 0_X)(0_X + 1_X)) = 0_X$. Suppose that a complemented domain γ separates the pair. Put $\Gamma = F\gamma$, $\Gamma' = F\gamma^\perp$, and $s = s_\gamma$. The domains Γ, Γ' on FX

are complementary, and $s : FX \to FX+FX$ is total. Write $j_0, j_1 : FX \to FX+FX$ for the target coproduct injections. Proposition 4.0.6 gives $s\Gamma = j_1\Gamma$ and $s\Gamma' = j_0\Gamma'$.

Let $P_0 = 1_{FX}+0_{FX}$ and $P_1 = 0_{FX}+1_{FX}$ be the summand domains on $FX+FX$. The morphisms $P_0 s$ and $j_0\Gamma'$ agree after both Γ and Γ'. Since $\mathrm{ran}[\Gamma, \Gamma'] = 1_{FX}$, range agreement gives $P_0 s = j_0\Gamma'$. Likewise $P_1 s = j_1\Gamma$. Taking restrictions gives

$$s^* P_0 = \Gamma', \qquad s^* P_1 = \Gamma.$$

Choose $d_\gamma : X \to X + X$ with $Fd_\gamma = \chi_{X,X} s$, and put $u = b^{-1} d_\gamma : X \to X$. Choose a total t-index $e : X \to X$ of u. Equation (2) gives $t\Delta_X e = ue$, and hence $he = d_\gamma e$. Put

$$H = \chi^{-1}_{X,X} Fh : FX \to FX + FX, \qquad E = Fe : FX \to FX.$$

Applying F to the preceding identity gives $HE = sE$. The canonical comparison satisfies $\chi_{X,X} P_0 = F(1_X + 0_X)\chi_{X,X}$ and $\chi_{X,X} P_1 = F(0_X + 1_X)\chi_{X,X}$. Since F preserves restrictions, it follows that $F\alpha = H^* P_0$ and $F\beta = H^* P_1$. Thus

$$E^* F\alpha = (HE)^* P_0 = (sE)^* P_0 = E^*\Gamma',$$
$$E^* F\beta = (HE)^* P_1 = (sE)^* P_1 = E^*\Gamma.$$

The inequalities $F\alpha \le \Gamma$ and $F\beta \le \Gamma'$ therefore give

$$E^*\Gamma' \le E^*\Gamma, \qquad E^*\Gamma \le E^*\Gamma'.$$

These two domains are equal, and their meet is $E^*(\Gamma\Gamma') = 0_{FX}$. Consequently both are zero. The morphism E is total, so the inverse-image join law gives

$$1_{FX} = E^* 1_{FX} = E^*(\Gamma \vee \Gamma') = E^*\Gamma \vee E^*\Gamma' = 0_{FX},$$

contrary to $1_X \notin \ker F$. □

The lifting hypothesis holds if $F : \mathbf{C}(X, X + X) \to \mathbf{D}(FX, F(X + X))$ is surjective. It also holds when every complemented $\gamma \in \mathrm{Dom}(X)$ has a total morphism $d_\gamma : X \to X + X$ with $d_\gamma\gamma = i_1\gamma$ and $d_\gamma\gamma^\perp = i_0\gamma^\perp$. Indeed, $p_\gamma d_\gamma = 1_X$, where $p_\gamma = \nabla_X(\gamma^\perp + \gamma)$: both sides agree after composition with γ and $\gamma^\perp$, and $\mathrm{ran}[\gamma, \gamma^\perp] = 1_X$. Thus $\chi^{-1}_{X,X} Fd_\gamma$ is a section of $\nabla_{FX}(F\gamma^\perp + F\gamma)$, and uniqueness gives $Fd_\gamma = \chi_{X,X} s_\gamma$.

Proposition 4.1.1. *Let* $\mathbf{C}$ *be a restriction category with restriction zeros and binary restriction coproducts. Suppose that every pair of parallel morphisms with disjoint domains has a join. For each complemented domain* $\gamma : X \to X$*, the morphism* $d_\gamma = (i_1\gamma) \vee (i_0\gamma^\perp) : X \to X + X$ *is total and satisfies* $d_\gamma\gamma = i_1\gamma$ *and* $d_\gamma\gamma^\perp = i_0\gamma^\perp$*. If* $F : \mathbf{C} \to \mathbf{D}$ *is a range functor and* $\mathbf{D}$ *satisfies the target hypotheses of Theorem 4.1, then* $Fd_\gamma = \chi_{X,X} s_\gamma$*, where* $s_\gamma : FX \to FX + FX$ *is the unique section of* $\nabla_{FX}(F\gamma^\perp + F\gamma)$*. Thus* $\chi_{X,X} s_\gamma$ *has a preimage under* F*, as required by the lifting hypothesis of Theorem 4.1.*

Proof. Since the coproduct injections are total, $\overline{i_1\gamma} = \gamma$ and $\overline{i_0\gamma^\perp} = \gamma^\perp$. The two morphisms therefore have disjoint domains. Their upper-bound equations give $d_\gamma\gamma = i_1\gamma$ and $d_\gamma\gamma^\perp = i_0\gamma^\perp$, and their domains satisfy $\gamma, \gamma^\perp \le \overline{d_\gamma}$. Hence $1_X = \gamma \vee \gamma^\perp \le \overline{d_\gamma}$, so d_γ is total. The section argument above gives $Fd_\gamma = \chi_{X,X} s_\gamma$. □

4.5. **Failure of the lifting hypothesis.** A target section $s_\gamma : FX \to FX + FX$ can have no source morphism $d_\gamma : X \to X + X$ with $Fd_\gamma = \chi_{X,X} s_\gamma$, even when F is a faithful range functor that preserves the near product and reflects zero morphisms. We give a counterexample in which the pair in Theorem 4.1 is separable.

Fix an acceptable enumeration (φ_e) of the partial recursive functions, and put $K = \{e : \varphi_e(e) \downarrow\}$. Let (W_a^K) be the enumeration of domains of an acceptable numbering of partial K-recursive functions. Fix a primitive recursive bijective pairing function with projections π_0, π_1.

The objects of $\mathbf{C}_K$ are the sets generated from $\mathbb{N}$ by binary products and tagged coproducts, with their recursive codings. Its morphisms $f : X \rightharpoonup Y$ are partial functions with K-computably enumerable domains that admit ordinary partial recursive extensions. Equality is equality of graphs, and composition is composition of partial functions. We write 1_B for the partial identity on a subset B and use $\simeq$ for equality of partial values, including simultaneous undefinedness. On $\mathbb{N}$, every morphism has the form

$$f_{e,a}(x) \simeq \begin{cases} \varphi_e(x), & x \in W_a^K, \\ \uparrow, & x \notin W_a^K. \end{cases}$$

The total morphisms are exactly the ordinary total recursive functions.

Proposition 4.1.2. *The category $\mathbf{C}_K$ is a P-recursion category with binary coproducts and ranges. Its inclusion $F : \mathbf{C}_K \to \mathbf{Pfn}$ satisfies all hypotheses of Theorem 4.1 except the lifting hypothesis; its diagonal pair can be recursively separable. Moreover, $\mathbf{C}_K$ contains two parallel morphisms with disjoint domains and no common upper bound in the restriction order.*

Proof. Write D_f for the domain of a partial function f, and let f_0, g_0 be partial recursive extensions of composable morphisms f, g. The composite has extension $g_0 f_0$ and domain $D_f \cap f_0^{-1}(D_g)$, which is K-computably enumerable. Identities belong to the category, and associativity holds for partial functions. Products, restrictions, and tagged copairings have the same closure property. Thus the set-theoretic near product and coproduct give the stated categorical structures. All objects are recursively isomorphic, and the empty partial functions give the prodominical zeros. The image of a K-computably enumerable set under a partial recursive function is K-computably enumerable. Its partial identity is the range, with the required equality property, and ranges commute with products.

Define $t(\langle e, a\rangle, x) = f_{e,a}(x)$. This morphism has a partial recursive extension and a K-computably enumerable domain. For $v : \mathbb{N} \times \mathbb{N} \rightharpoonup \mathbb{N}$, choose a partial recursive extension $V(z, x)$, and write D_v for the K-computably enumerable domain of v. The ordinary and oracle parameterization theorems [Odi89] give ordinary total recursive functions $e(z), a(z)$ such that $\varphi_{e(z)}(x) \simeq V(z, x)$ and $W_{a(z)}^K = \{x : (z, x) \in D_v\}$. Oracle parameterization substitutes z into program code and is ordinary recursive; the oracle is used by the indexed program. Consequently $c(z) = \langle e(z), a(z)\rangle$ is total in $\mathbf{C}_K$ and $v = t(c \times 1)$. Recursive object codings handle the other parameter objects, proving the Turing property.

The inclusion into **Pfn** preserves restrictions, ranges, coproducts, zeros, and near products. The target is split and locally connected, its inverse images preserve joins, and injective partial functions have partial inverses, so weak choice holds.

Take $b(2n) = i_0(n)$ and $b(2n+1) = i_1(n)$, and put $p(q) \simeq \varphi_{\pi_0(q)}(q)$. The diagonal domains are

$$D_\alpha = \{q : q \in W^K_{\pi_1(q)},\ p(q)\downarrow \text{ with an even value}\},$$
$$D_\beta = \{q : q \in W^K_{\pi_1(q)},\ p(q)\downarrow \text{ with an odd value}\}.$$

The set $A = \{q : p(q)\downarrow$ with an even value$\}$ is computably enumerable, hence decidable relative to K. Both A and A^c are K-computably enumerable, so their partial identities are complementary domains of $\mathbf{C}_K$. Since $D_\alpha \subseteq A$ and $D_\beta \subseteq A^c$, the pair is separable.

To identify the failed lift, fix an index a_0 and choose $e(n)$ effectively for a program that ignores its input, simulates $\varphi_n(n)$, and returns 0 on convergence. Then $n \in K$ if and only if $\langle e(n), a_0\rangle \in A$, so A is nonrecursive. The morphism $p_A = \nabla(1_{A^c} + 1_A) : \mathbb{N} + \mathbb{N} \rightharpoonup \mathbb{N}$ has a unique section in **Pfn**, given by

$$s_A(n) = \begin{cases} i_1(n), & n \in A, \\ i_0(n), & n \notin A. \end{cases}$$

This function and $b^{-1}s_A$ are total and nonrecursive. They therefore have no preimages under the inclusion F.

The morphisms $f = i_1 1_A$ and $g = i_0 1_{A^c}$ from $\mathbb{N}$ to $\mathbb{N} + \mathbb{N}$ belong to $\mathbf{C}_K$ and have disjoint domains. If d were a common upper bound, then $d1_A = f$ and $d1_{A^c} = g$. Since $A \cup A^c = \mathbb{N}$, these equations force $d = s_A$, contradicting the fact that every total morphism of $\mathbf{C}_K$ is recursive. Thus the joins required by Proposition 4.1.1 fail to exist, although the domains of $\mathbf{C}_K$ have binary joins. □

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